Gaze behaviour reveals how cross-linguistic differences in case alignment shape the dynamics of sentence planning

The time course of sentence planning can vary in response to, for example, extralinguistic factors such as time pressure[1] or the visual salience of referents[2]. In recent years, evidence has accumulated that the grammatical properties of a language can exert a strong influence on sentence production processes[3–5]. It remains largely unknown, however, whether adaptations to grammatical affordances are the same across languages, mainly due to a lack of comparative cross-linguistic production research[5,6]. Here, we extend the empirical basis of sentence production research by investigating how differences in case marking alignment affect the time course of planning in Yélî Dnye (a language isolate of Papua New Guinea) and Japanese.

While nominative-accusative languages use the same case (the nominative) to mark the agents of both transitive and intransitive verbs, ergative languages use a dedicated case (the ergative) to exclusively mark the agents of transitive verbs. Previous studies on Hindi and Basque found that planning sentences with ergative case alignment induces earlier and more extensive relational encoding at the outset of formulation because speakers must determine the "who-does-what-to-whom" structure of the event early to select the appropriate case marker on the initial argument[7,8]. However, in those languages, ergative case marking is conditioned by language-specific grammatical factors beyond syntactic transitivity (grammatical aspect in Hindi and lexical specification in Basque), leaving it unresolved whether it is ergativity itself which drives earlier relational encoding or another co-dependent grammatical property of the languages tested so far.

We conducted an eye-tracked picture description task with speakers of Yélî Dnye, a language with strictly ergative case marking (n = 45), and speakers of Japanese, a language with nominative-accusative case marking (n = 56). The default word order in both languages is Agent-Patient-Verb (APV). Speakers of the two languages described 52 colored cartoon scenes depicting transitive (two-participant) events while their eye movements were recorded (60 Hz sampling rate). Binomial Bayesian mixed-effects growth curve regressions modeled the time course of visual attention to the agent and patient characters in the scenes, controlling for a range of nuisance predictors. In an early time window (200-800 ms), associated primarily with conceptual encoding, Yélî Dnye speakers prioritised relational encoding to a greater degree than Japanese speakers. This is evidenced by Yélî Dnye speakers distributing their attention more between the depicted characters, leading to fewer agent and more patient fixations than in Japanese speakers (Figure 1). Differences between the languages are reflected in improved predictive performance in 10-fold leave-participants-out cross-validation when models included a language predictor and its interactions with the polynomial time terms (i.e., $\Delta_{eldp} > 2 \times SE_{eldp}$, for both agent and patient fixations[9]), and by predictors for language×time interactions exceeding posterior probabilities > 80%, with < 5% of the posterior probability mass falling within a region of practical equivalence (ROPE) of ±0.06 logits[10].

These results support the conclusion that it is ergative case marking per se that consistently induces extensive early relational coding (in Yélî Dnye, but also in Hindi[8] and Basque[7,11]). Thus, across diverse languages, the uptake of visual information relevant for utterance preparation is shaped by the affordances imposed by a language's grammar from the earliest stages of sentence planning.

Although the Papua and New Guinea area is home to a fifth of the world's languages[12], currently only a single published study on sentence processing in a Papuan language exists[4]. Our study thus also provides a window into the psycholinguistics of languages of the most linguistically diverse region of the world, broadening our understanding of how typological variation shapes the time course of language processing.

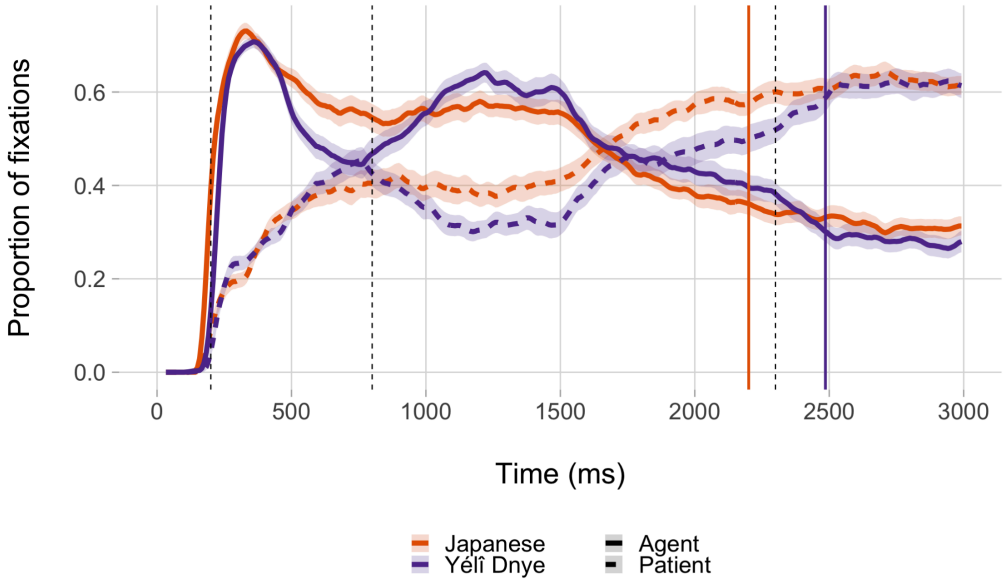

Figure 1. Grand mean proportions of Yélî Dnye and Japanese speakers' agent and patient fixations over time. Ribbons indicate one standard error of the mean. The dotted vertical lines indicate the analysis time windows (200-800 and 800-2300 ms after stimulus onset); the solid vertical lines show the grand mean speech onsets for each language.

References

1. Ferreira, F. & Swets, B. How incremental is language production? Evidence from the production of utterances requiring the computation of arithmetic sums. *J. Mem. Lang.* **46**, 57–84 (2002).
2. Myachykov, A., Thompson, D., Garrod, S. & Scheepers, C. Referential and visual cues to structural choice in visually situated sentence production. *Front. Psychol.* **2**, 396 (2011).
3. Nordlinger, R., Garrido Rodriguez, G. & Kidd, E. Sentence planning and production in Murrinhpatha, an Australian "free word order" language. *Language* **98**, 187–220 (2022).
4. Sarvasy, H. S., Morgan, A. M., Yu, J., Ferreira, V. S. & Momma, S. Cross-clause planning in Nungon (Papua New Guinea): Eye-tracking evidence. *Mem. Cognit.* **51**, 666–680 (2023).
5. Jaeger, T. F. & Norcliffe, E. J. The cross-linguistic study of sentence production. *Lang. Linguist. Compass* **3**, 866–887 (2009).
6. Norcliffe, E., Harris, A. C. & Jaeger, T. F. Cross-linguistic psycholinguistics and its critical role in theory development: early beginnings and recent advances. *Language, Cognition and Neuroscience* **30**, 1009–1032 (2015).
7. Egurtzegi, A. *et al.* Cross-linguistic differences in case marking shape neural power dynamics and gaze behavior during sentence planning. *Brain Lang.* **230**, 105127 (2022).
8. Sauppe, S. *et al.* Neural signatures of syntactic variation in speech planning. *PLoS Biol.* **19**, e3001038 (2021).
9. Yates, L. A., Aandahl, Z., Richards, S. A. & Brook, B. W. Cross validation for model selection: A review with examples from ecology. *Ecol. Monogr.* **93**, (2023).
10. Kruschke, J. K. Rejecting or accepting parameter values in Bayesian estimation. *Adv. Methods Pract. Psychol. Sci.* **1**, 270–280 (2018).
11. Isasi-Isasmendi, A. *et al.* The Agent Preference in Visual Event Apprehension. *Open Mind* **7**, 240–282 (2023).
12. *The Languages and Linguistics of the New Guinea Area*. (De Gruyter Mouton, Berlin, Germany, 2018).